InSoLiTo project: The research software graph-based network from OpenEBench

Sergi Aguiló-Castillo[1,2], José M. Fernández[1,2], Josep Ll. Gelpí[1,2,3] and Salvador Capella-Gutiérrez[1,2]
[1] Spanish National Bioinformatics Institute (INB/ELIXIR-ES)
[2] Barcelona Supercomputing Center (BSC)
[3] Dept Biochemistry and Molecular Biomedicine, Universitat de Barcelona (UB)

OPENEBENCH

InSoLiTo
(Infering)->(Social Life)<-[of]-(Tools)

InSoLiTo is a graph-based network of the co-usage of research software from the bioinformatics tools collection hosted by OpenEBench. Co-usage is understood as being cited in the same scientific publications. The initial aim of the project is to identify **how bioinformatics software relate to each other** allowing to **potentially infer analytical workflows** commonly used from the literature.

The idea

Is there any social network to know the closest friends of my bioinformatics tools?

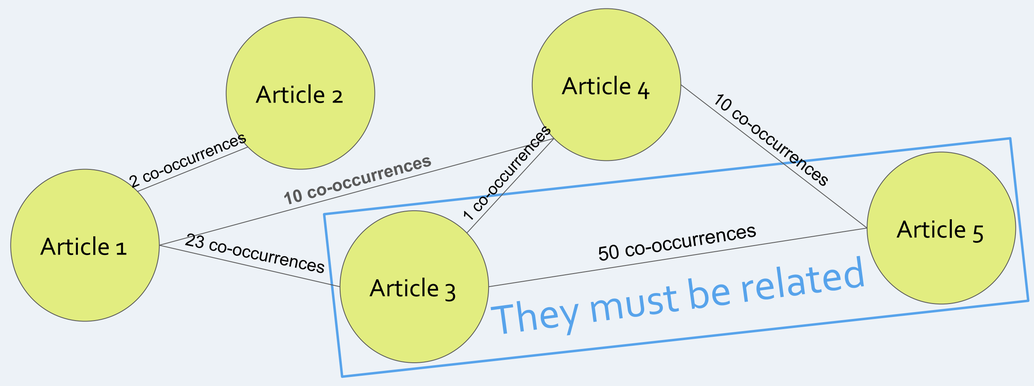
No, but that's a great idea! This way, you would know what other software your tool always work with.

To know how the bioinformatics community use their different software together, I think the best way is just to look to scientific literature.

Yes! And we create a new relationship in our social network every time we see two software working together in an scientific article.

Calculating the Co-occurrences

How many scientific publications two articles have been referenced together.

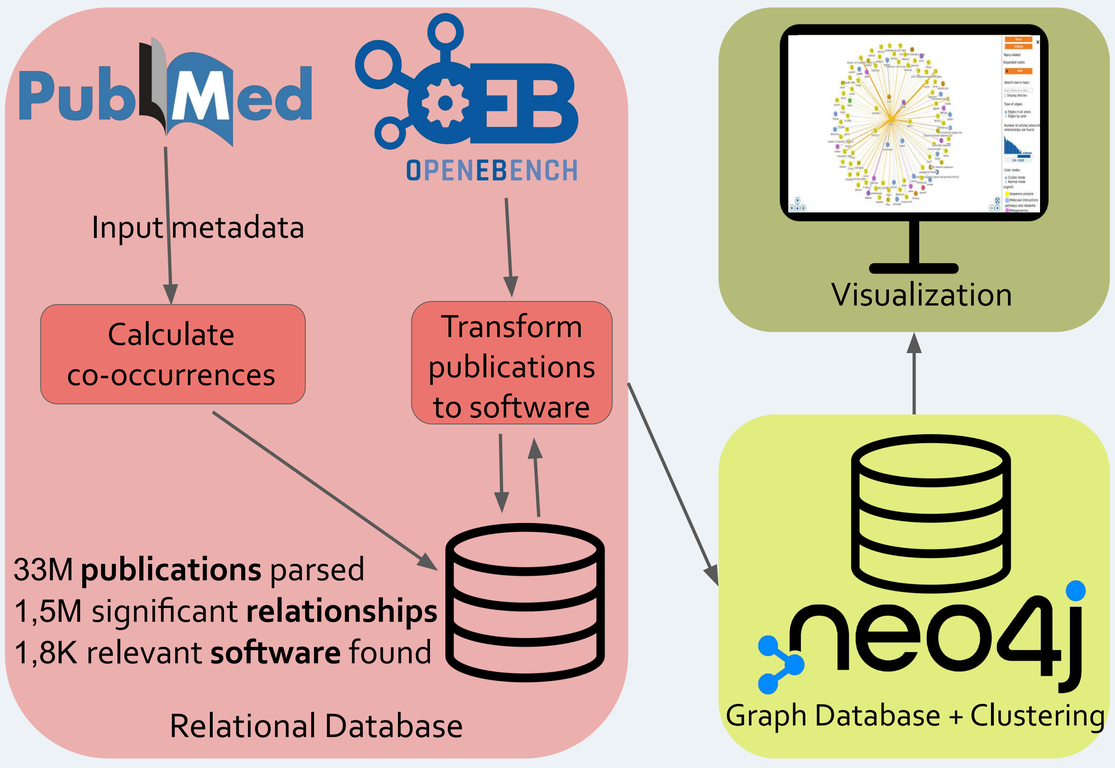


The most direct way of knowing how the bioinformatics community uses their software is by looking to the **references** in their scientific articles.
The theory is simple:
Two articles are closely related when there is high co-occurrence between them.

But, the people do not cite the software! They cite the publications associated to the software. How do we deal with that?

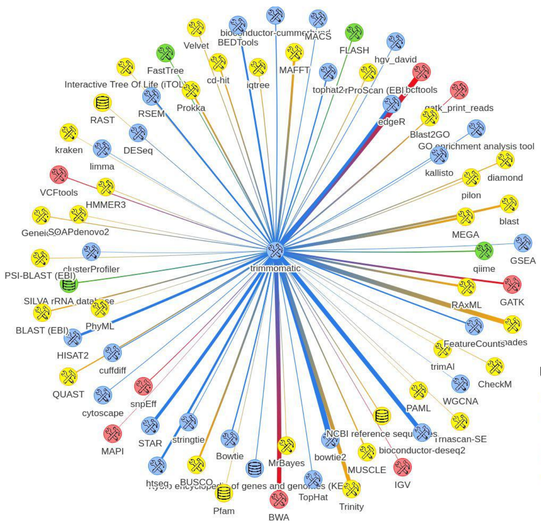
Don't worry, OpenEBench stores this kind of association. And much information that we can use for enriching our social network, such as EDAM ontology terms!

Development



Let's play!

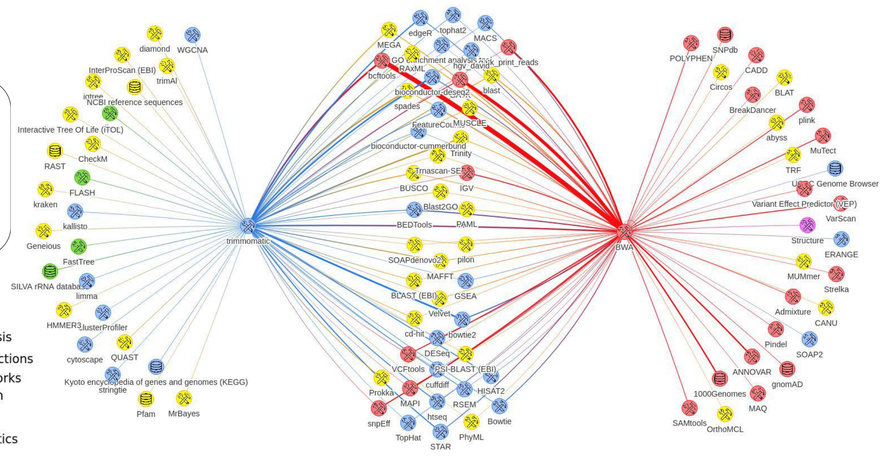
Who are the best friends of Trimmomatic?



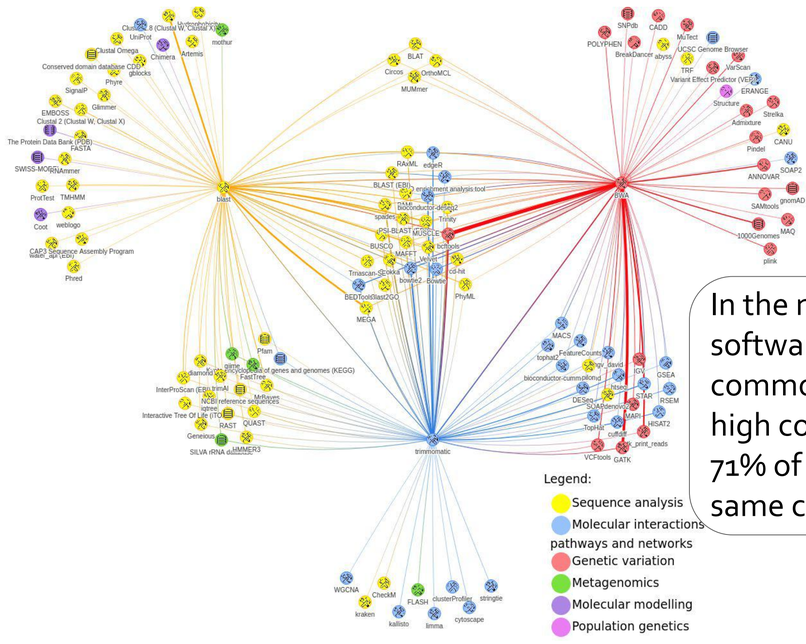
Filter by Trimmomatic with >200 co-occurrences in its relationships

Nodes colored by cluster

And what are the software that are close friends of Trimmomatic and BWA?

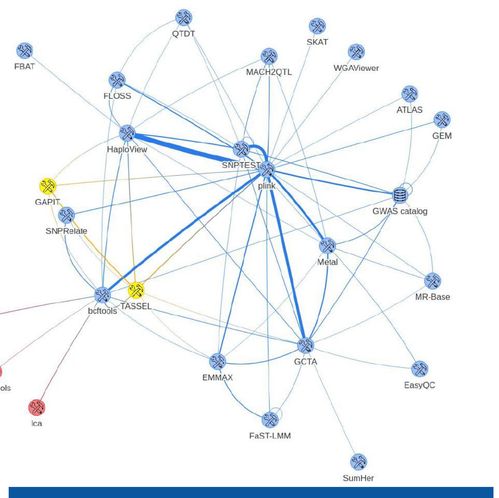


And what happens if you add Blast?

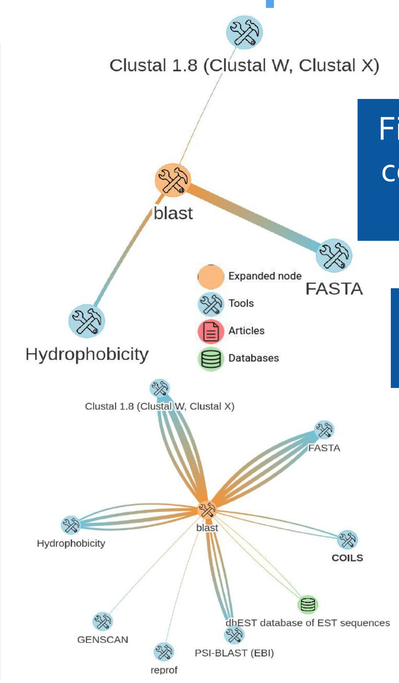


In the network, you see who are the software the three tools work in common. Moreover, clusters show a high correlation with EDAM Topic (Ex: 71% of Proteomics software is in the same cluster).

And there are other options



Filter by bioinformatics domain
Ex: GWAS Study



Filter by Blast and co-occurrences in 1995

Nodes colored by type of node

Filter by Blast and co-occurrences from 1995 to 2000

Scan this QR to play with the social network:

Sergi Aguiló-Castillo
Junior Research Engineer
INB/ELIXIR-ES
sergi.aguilo@bsc.es

Spanish National Bioinformatics Institute

elixir SPAIN

Barcelona Supercomputing Center
Centro Nacional de Supercomputación

insolito.openebench.bsc.es